

Week 3 Update — COVID-19 Global Data Analysis

Shrey Jain • Top 10 countries by total cases & fatality rate • Prepared for Neelam Gupta

Weekly update

This week I ranked countries by total confirmed cases, took the top 10, and compared them on fatality rate (naive CFR = deaths / cases) and deaths per million. Analysis uses country-level rows only (aggregates such as *World* and income groups are excluded) and each country's latest record, in a reproducible Plotly notebook.

Discovered:

- **Fatality rate varies widely** across the top-10-by-cases group — from ~0.10% (South Korea) and ~0.12% (China) up to ~1.87% (Brazil) and ~1.18% (India).
- **High case counts do not imply high fatality.** Several of the largest-case countries (South Korea, Japan, China) sit at the low end of CFR.
- On **deaths per million**, the United States (~3,490) and United Kingdom (~3,400) lead the group; China (~86) and India (~374) are lowest.
- **Brazil** is the notable case: large case count and the highest CFR in the group.

Top 10 countries by total cases

#	Country	Total cases	Total deaths	CFR	Deaths / M
1	United States	103,436,829	1,193,165	1.15%	3,494
2	China	99,373,219	122,304	0.12%	86
3	India	45,041,748	533,623	1.18%	374
4	France	38,997,490	168,091	0.43%	2,615
5	Germany	38,437,756	174,979	0.46%	2,081
6	Brazil	37,511,921	702,116	1.87%	3,339
7	South Korea	34,571,873	35,934	0.10%	694
8	Japan	33,803,572	74,694	0.22%	598
9	Italy	26,781,078	197,307	0.74%	3,309
10	United Kingdom	24,974,629	232,112	0.93%	3,404

Comparison charts

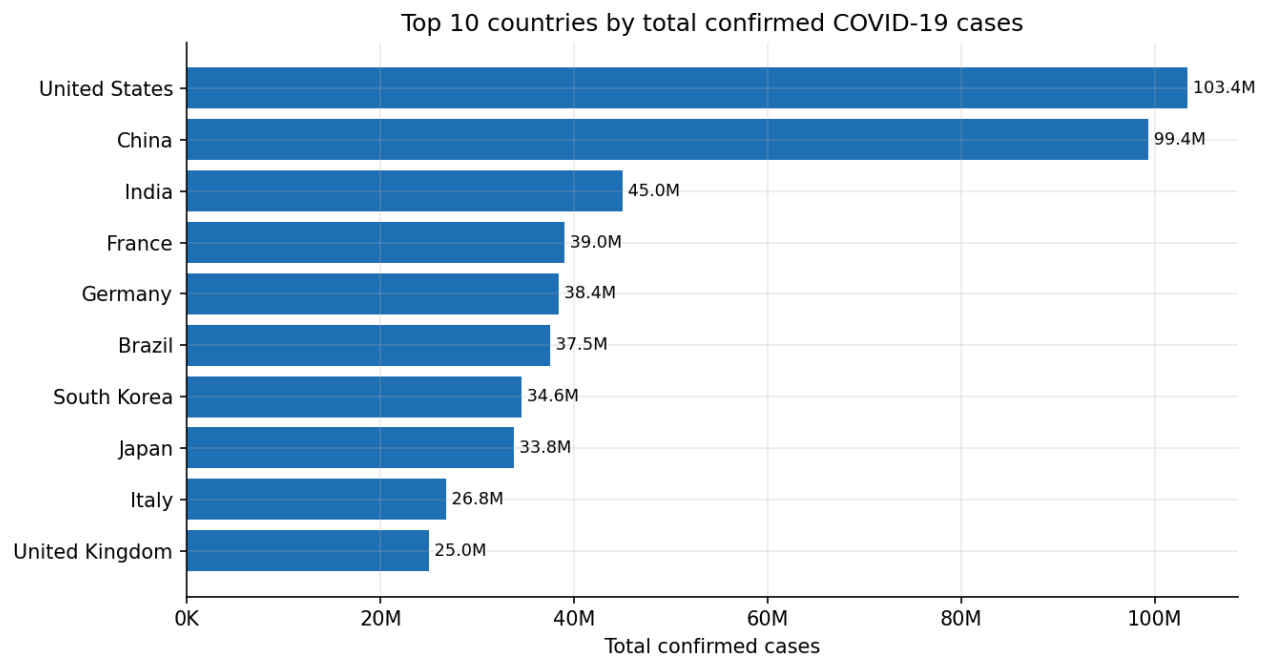


Fig 1. Top 10 countries by total confirmed cases.

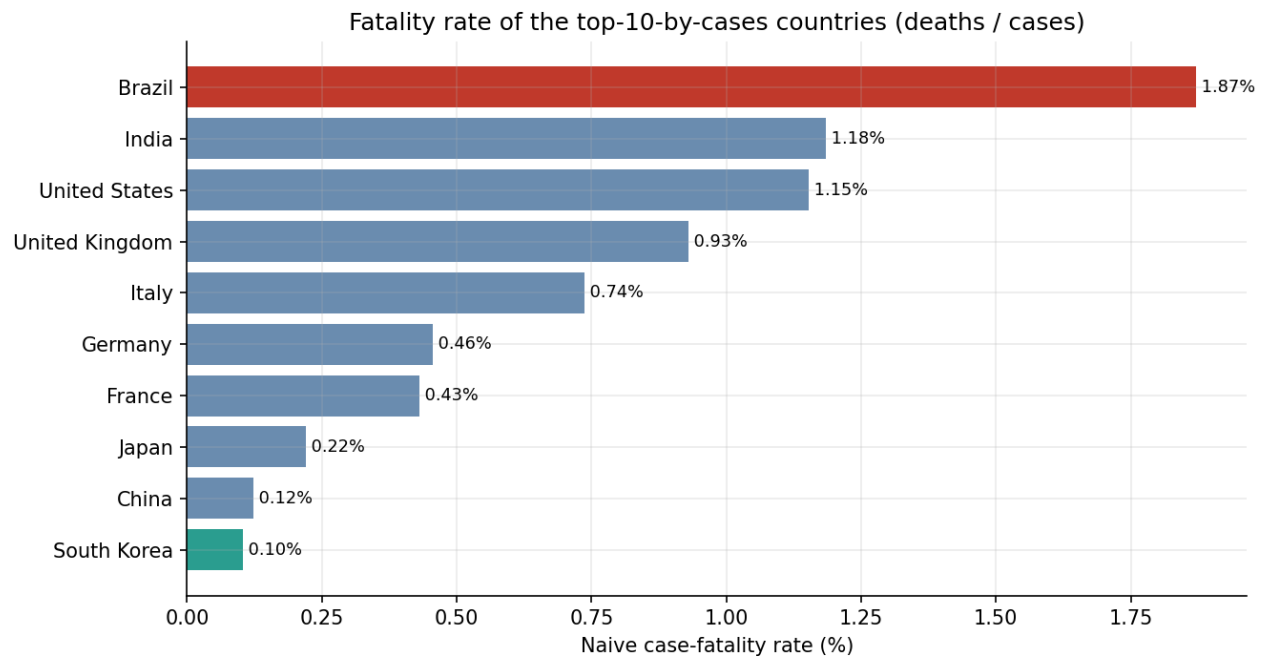


Fig 2. Naive fatality rate (CFR) for the same 10 countries, sorted. Lowest in teal (South Korea), highest in red (Brazil).

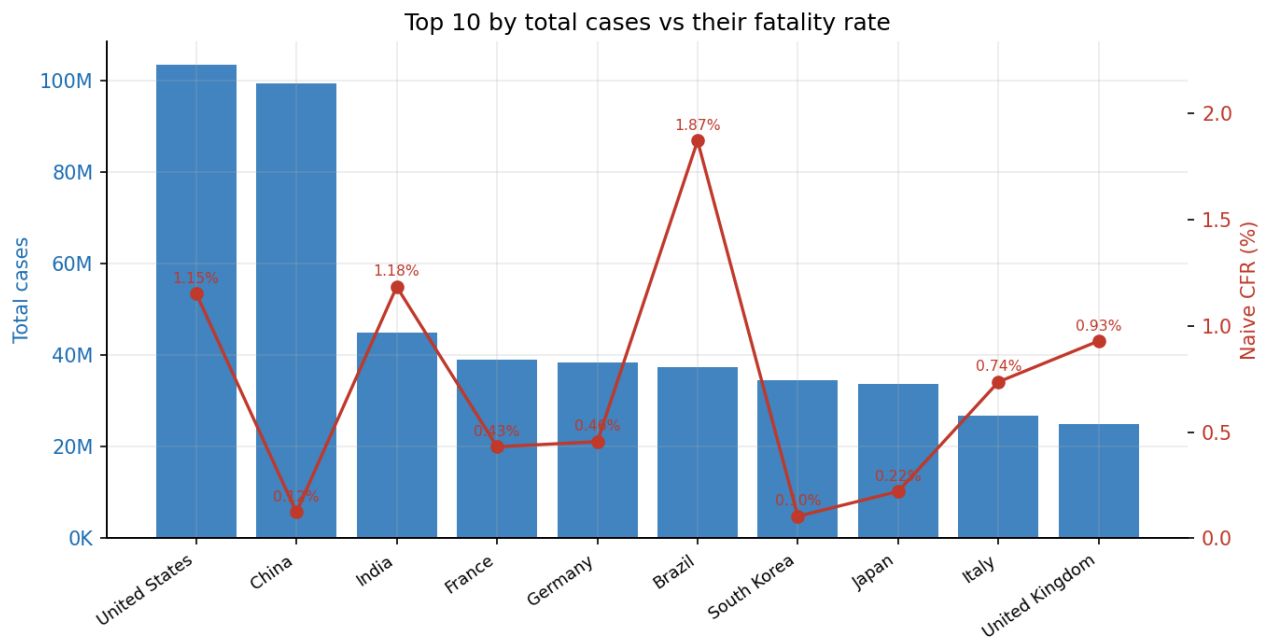


Fig 3. Direct comparison: total cases (bars) vs fatality rate (red markers). The two do not track each other.

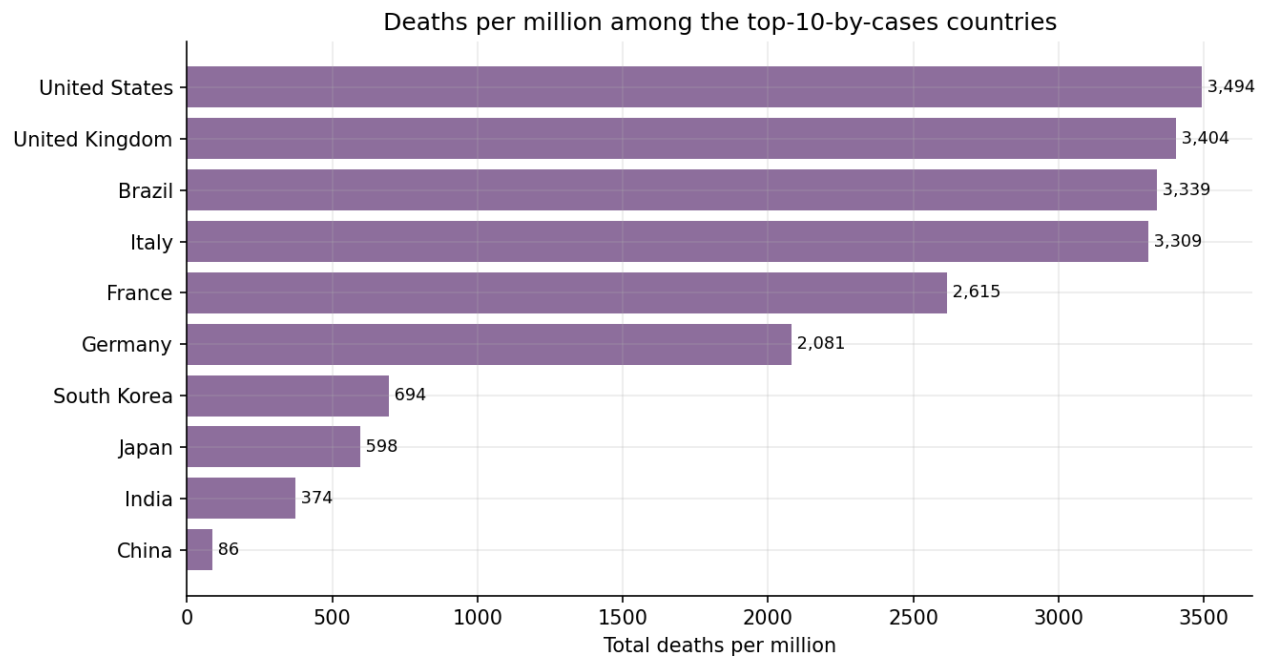


Fig 4. Deaths per million — a population-adjusted mortality view for the same group.

Work artifacts & data sources

Repository	github.com/shreyjain99/covid19-global-analysis
Week 3 folder	github.com/shreyjain99/covid19-global-analysis/tree/main/week%203
Week 3 notebook	week3_top10_cases_cfr.ipynb (interactive Plotly)

Charts	figures/fig1–fig4 (.png) + top10_table.csv
Data source	Our World in Data COVID-19 dataset (github.com/owid/covid-19-data)

Reproducibility (how to re-run)

- `pip install -r requirements.txt`.
 - Open `week3_top10_cases_cfr.ipynb` and Run All — it downloads the dataset automatically and regenerates every figure from the raw data.
 - Or from the command line: `python scripts/analysis_week3.py (charts)` then `python scripts/build_week3_report.py` (this PDF).
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Risks / blockers & support needed

- **No blockers to delivery.** The week's objective is met and reproducible.
- **Measurement caveat:** naive CFR and reported totals depend on each country's testing and death-reporting practices, so cross-country CFR gaps partly reflect measurement rather than lethality (China's reported figures in particular are widely regarded as undercounts).
- **Support:** confirm whether reported CFR is acceptable for this comparison, or whether an adjusted / excess-mortality basis is preferred.